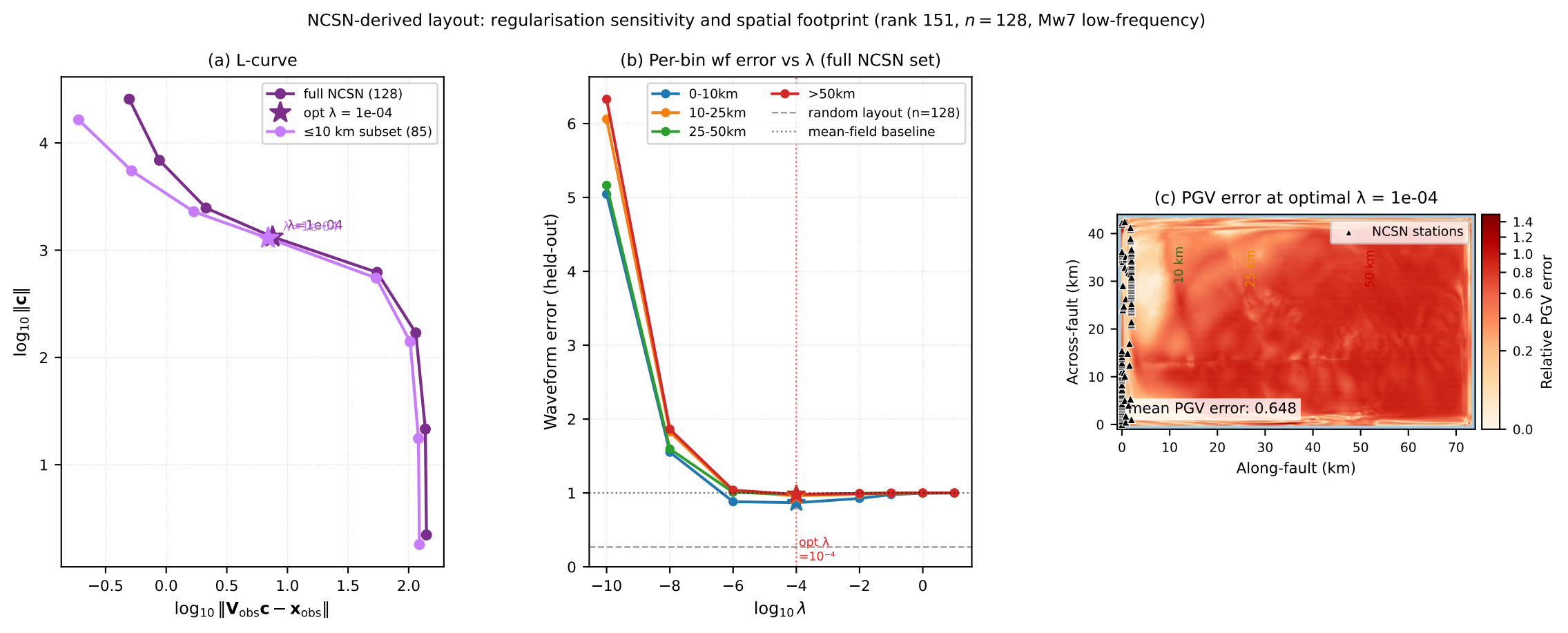

NCSN-derived layout: regularisation sensitivity and spatial footprint (rank 151, n = 128, Mw7 low-frequency)
(a) L-curve
full NCSN (128)
opt λ = 1e-04
≤10 km subset (85)
λ=1e-04
log10 ‖c‖
log10 ‖Vobs c − xobs‖
(b) Per-bin wf error vs λ (full NCSN set)
0-10km
10-25km
25-50km
>50km
random layout (n=128)
mean-field baseline
opt λ =10⁻⁴
Waveform error (held-out)
log10 λ
(c) PGV error at optimal λ = 1e-04
NCSN stations
10 km
25 km
50 km
mean PGV error: 0.648
Across-fault (km)
Along-fault (km)
Relative PGV error